

THE EXPRESSION OF SELF-INCOMPATIBILITY IN ANGIOSPERMS IS BIMODAL

Andrew R. Raduski,^{1,2} Elizabeth B. Haney,¹ and Boris Igić¹

¹*Department of Biological Sciences, University of Illinois at Chicago, 840 West Taylor Street, M/C 067, Chicago, Illinois 60607*

²*E-mail: aradus2@uic.edu*

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Self-incompatibility is expressed by nearly one-half of all angiosperms. A large proportion of the remaining species are self-compatible, and they either outcross using various contrivances or self-fertilize to some extent. Because of the common occurrence of populations and individuals with intermediate levels of self-incompatibility, categorization of the expression of self-incompatibility as an approximately binary trait has become controversial. We collect a widely reported index (index of self-incompatibility [ISI]) used to assess the strength and variation of self-incompatibility from over 1200 angiosperm taxa. Its distribution is bimodal and positively associated with outcrossing rate, albeit with a weak relationship within self-compatible taxa. A substantial fraction of species has intermediate mean values of ISI. Their occurrence can be caused by segregating ephemeral self-compatible mutations, averaging artifacts, and experimental biases, in addition to the often invoked stabilizing selection acting on the expression of self-incompatibility. Selection may also generally favor taxa with high ISI values through increased lineage birth and death rates, and it may counter lower level selection advantages within taxa expressing intermediate and low values of ISI. Such a null hypothesis is nearly universally overlooked, despite the fact that it could adequately explain the observed distribution of mating and breeding systems.

KEY WORDS: Breeding systems, levels of selection, mating systems, outcrossing rate, self-compatibility.

One of the distinguishing features of flowering plants is their outstanding propensity for variation in breeding systems—the collection of physiological and morphological traits that determine the likelihood that any two gametes will unite. The study of mating systems, or realized modes of reproduction, has dominated the literature on plant evolution for decades (Fryxell 1957; Barrett 2003). A great deal of attention turned to the empirical distribution of mating systems, the associated population genetic and ecological parameters, and increasingly complex theoretical models aimed at recovering stable mixed mating strategies (Barrett 2003; Goodwillie et al. 2005). It is generally thought that lower level selection drives the evolution of mating systems (phenotypes are expressed at the individual level, and optimal values depend on values of other individuals in a population), through many factors, including the automatic advantage of selfing, reproductive assurance, sex allocation, and a dynamic influence on the

evolution of inbreeding depression (Lloyd 1979; Charnov 1982; Lande and Schemske 1985; Uyenoyama 1986; Goodwillie et al. 2005; Johnston et al. 2009). At the same time, higher level selection may affect the rate of speciation and extinction associated with different aggregate values of breeding and mating systems in populations and species (Dobzhansky 1950; Stebbins 1957; Williams 1992; Holsinger 2000; Takebayashi and Morrell 2001; Igić et al. 2004). Therefore, a strategy that is advantageous in the short-term may be associated with a lower net diversification rate, and commit a lineage to a suboptimal long-term path.

Even the earliest studies of plant breeding systems noted that despite important potential advantages of traits associated with self-fertilization, many flowering plant species rely on intricate adaptations used for outcrossing (Darwin 1876). The most common and phylogenetically widespread mechanism for avoidance of self-fertilization in flowering plants is self-incompatibility (SI),

the inability of otherwise fertile co-sexual plants to set seed when self-pollinated, which employs a genetic pathway to recognize and reject a plant's own pollen (de Nettancourt 1977). Approximately 50% of angiosperm species from over 100 plant families use one of many convergently evolved SI mechanisms (Igić et al. 2008). Independent lines of evidence indicate that the shift from SI to self-compatibility (SC) is one of the most frequent evolutionary transitions in flowering plants (Stebbins 1974; Igić et al. 2008). Such a shift clearly requires hereditary individual-level variation in SI response, and there is little doubt that any closely studied SI populations or species will reveal a continuously varying distribution of floral traits as well as SI, which cannot be classified easily (East 1940; Ascher and Peloquin 1966; Darwin 1876; Stephenson and Bertin 1983; Levin 1996; Lipow et al. 1999; Stephenson et al. 2000; Goodwillie et al. 2004; Ferrer et al. 2009). Lloyd (1965) formally introduced the most widely reported quantitative metric of SI, the index of self-incompatibility (ISI), a relative measure of reproductive success following hand self- and cross-pollinations in his treatise on *Leavenworthia* (Brassicaceae). The importance of such a metric for the strength of SI, a proxy for the outcrossing propensity of individuals, was realized much earlier. It did not escape Darwin (1878, Chapter IX), but Lloyd's formulation allows easy comparative study across taxa. Many studies reported breeding systems of hermaphrodite species as qualitative characters, and comparative studies to date almost exclusively rely on discrete, often binary, classifications (Goodwillie et al. 2005).

There is a growing discontent with the binary SI versus SC classification, often criticized for its inability to capture the apparent variety of rule-breaking intermediate states between SI and SC (Stephenson et al. 2000; Goodwillie et al. 2004; Ferrer et al. 2009). The increase in popularity of this viewpoint is spurred by widespread reports of variation in the strength of SI within and among individuals and populations, attributed to both genetic and environmental factors, as well as the contention that the observed variation may be maintained by stabilizing selection over evolutionary time (Goodwillie et al. 2004; Vallejo-Marin and Uyenoyama 2004; Ferrer and Good-Avila 2007). For over a century, virtually all researchers and influential monographs on the subject explicitly acknowledged variation in SI response but then heuristically classified populations or species, frequently as either SI or SC. This practice continues in part because many statistical phylogenetic methods require discrete data (e.g., Igić et al. 2004; Ferrer and Good-Avila 2007). The magnitude and direction of errors and biases resulting from qualitative assignments should depend on how well a binary classification approximates the empirical distribution of the strength of incompatibility reactions.

We use data collected from the literature on plant breeding systems to estimate ISI distribution. Although the expected shape of the ISI distribution is unclear, the history of binary SI classification predicts a bimodal distribution. But if stabilizing selection on

the expression of SI at the lower levels of evolutionary hierarchy operates pervasively and strongly in flowering plants, intermediate values of ISI should be the most frequent. Alternatively, two other hypotheses are commonly cited (none are mutually exclusive over evolutionary time). First, disruptive selection at lower levels may have a strong hand in shaping the distribution (Lande and Schemske 1985). Alternately, if species with low or intermediate ISI have a reduced net diversification rate they may not persist over long time periods, as previous results have suggested (Igić et al. 2004, 2006; Goldberg et al. 2010). Also, the strength of ISI is often used as a proxy for outcrossing and we examine the relationship between the estimates of ISI and outcrossing rate (t), a measure of mating system. Most fertilization events involving individuals with high ISI should occur by outcrossing. On the other hand, absence of SI should not necessitate low outcrossing rates, because a variety of alternative mechanisms such as dichogamy or herkogamy, combined with small floral display sizes, can also increase the likelihood of outcrossing. Therefore, we expect ISI to have a potentially complex relationship with outcrossing rates. High ISI should be strongly causally associated with high outcrossing rates, but low ISI may be only weakly associated with outcrossing rates. Species in the ISI dataset may be nonrandomly sampled at least with respect to their phylogenetic placement, geography, and life-history composition, and the shape of the distribution may be shaped by high mutation rates, averaging, study bias, as well as experimental error. Below, we present a set of hypotheses and a dataset, presumably containing some valuable information about plant breeding systems, which may together spur further interest in the empirical and theoretical study of breeding systems.

Methods

We collected published reports from which relative success of selfed and outcrossed hand-pollinations could be calculated, or those that directly provided a measure along with transparent calculation procedures by searching the online databases ISI Web of Knowledge Science Citation IndexTM and Google Scholar (<http://www.google.com/schhp>), using dozens of search terms related to plant mating and breeding systems. Our search included papers published through May 2011. In addition, we augmented the dataset with a personal library (B.I.) of books, monographs, and manuscripts, ranging from about 1850 through 1995. The complete list of references is listed in Supporting information.

Following Lloyd (1965), we measure ISI as

$$ISI = 1 - \frac{\text{relative selfed success}}{\text{relative outcrossed success}},$$

where relative pollination success is defined as the ratio of fruits set to flowers pollinated. It is possible for ISI to take on negative

values in the rare instances when the relative success of self-pollinations is higher than that of cross-pollinations. The preponderance of negative ISI values probably result from statistical and experimental sources of error (low sample sizes, incompatibility of outcross pollen, mistakes). That is, there are meager biological grounds for higher fruit set after self-pollination than after cross-pollination within a single population (Lloyd and Schoen 1992). Consequently, we set all negative values of ISI to zero ($n = 150$). Historically, species with ISI values above 0.8 have been classified as SI (Bawa 1974).

Unless noted otherwise, we use two-sample nonparametric Kolmogorov–Smirnov tests for all comparisons involving ISI and t distributions. The Kolmogorov–Smirnov test does not make any assumptions about the shape of the underlying distribution or the distribution of sampling error, so long as the distribution is continuous (Young 1977). We apply a study-wide Holm–Bonferroni correction to all reported statistical tests. The measures of ISI are not solely determined by SI-related phenomena. For example, agamospermy (asexual production of seed), seedless fruit set, early-acting inbreeding depression, and experimental errors and biases may affect average ISI values. The occurrence of each of these may be relatively rare or may generate errors that generally result in apparently reduced bimodality.

Presently, we use and discuss only the ISI data based on fruit set per flower, because of the unknown frequency and magnitude of inbreeding depression and other phenomena on seeds set (see Supporting information). We are not aware of any evidence that suggests a measurable effect of inbreeding depression on fruit set. What is perhaps the most comprehensive study of inbreeding depression did not assess its effect on fruit set (Husband and Schemske 1996). Despite this potentially confounding effect, ISI data for species with fruit types that make separate scoring of fruit set difficult (e.g., achenes in Asteraceae) are included. We were able to partly correct for the occurrence of agamospermy, by subtracting the proportion of seed set—in studies that controlled for its occurrence—from both self- and cross-pollination treatments. The use of different individual measures of ISI (seed- vs. fruit-based measures of relative success), or correction for agamospermy, did not qualitatively affect the results we present below (see Supporting information).

To determine the correlation between ISI and t , we sought taxa in which estimates of both parameters were available. The mean outcrossing rates were obtained from the databases used by Igić and Kohn (2006) and Barringer (2007) supplemented with recently published reports. Outcrossing rates reported as greater than one ($t > 1$) were collapsed to one ($t = 1$; $n = 5$) because it is unlikely that the marker loci used for outcrossing rate calculation are driving disassortative mating (Schemske and Lande 1985; Igić and Kohn 2006). When multiple outcrossing rate estimates were reported for a single species, all available estimates were averaged.

Results

Our literature search of experimental crossing data yielded a measure of ISI for 1238 species in 144 plant families. The frequency distribution appears strongly bimodal (Fig. 1). In fact, 74.5% of species take on values $ISI \leq 0.2$ or $ISI \geq 0.8$. We find that 43.9% of species are classified as SI, using Bawa's cutoff value of $ISI \geq 0.8$. A large proportion of reported values (45.2%) were from 10 families: Fabaceae ($n = 117$), Solanaceae (94), Bromeliaceae (82), Orchidaceae (54), Rubiaceae (39), Proteaceae (38), Asteraceae (35), Stylidiaceae (34), Rosaceae (34), and Ericaceae (32). The ISI distributions within the most common families and the entire dataset, excluding the most common families, are also bimodal (see Supporting information). When the dataset was divided into species alternately classified as either woody or herbaceous, perennial or annual, and anemophilous or zoophilous, those distributions were individually also either bimodal or unimodal around an extreme value (see Supporting information). A subset of studies reported ISI values for individual plants. Considerable variation is found in the number of individuals reported, as well as the frequency distributions from these reports (Fig. 1). The mode of each species distribution is most often found at $ISI = 0$ or $ISI = 1$ with relatively few individuals showing intermediate ISI values.

Both ISI and t data were available for 70 species in 32 families. In 17 particularly valuable instances, the measurements were made in the same populations. Overall, the correlation between ISI and t is positive ($r^2 = 0.27$, $P < 0.001$). This t distribution is not significantly different than a previously reported one (Igić and Kohn 2006; $D = 0.164$, $P > 0.38$), which was corrected for an apparent bias against obligately outcrossing taxa. We find that taxa with $ISI \geq 0.8$ have a significantly lower variance in t than species with $ISI \leq 0.2$ (Fig. 2; Fligner–Killeen test: med $\chi^2 = 11.52$, $P < 0.05$).

Discussion

It is widely agreed in the botanical literature that despite the traditional presence–absence scoring of SI, the occurrence of species with intermediate average values of ISI is common (Lloyd and Schoen 1992; Levin 1996; Lipow et al. 1999; Stephenson et al. 2000; Nielsen et al. 2003; Goodwillie et al. 2005; Mena-Ali and Stephenson 2007). Nevertheless, there have been remarkably few attempts to systematically collect the data on ISI across species and estimate its distribution (but see Kenrick and Knox 1989; Burbidge and James 1991; Lloyd and Schoen 1992; Bullock 1995). Over 30% of species are aggregated at the low end of the distribution, with $ISI \leq 0.2$, 25.5% are found within $0.2 < ISI < 0.8$, and 43.9% have $ISI \geq 0.8$ (Fig. 1). The last category yields an approximation of the total number of SI species. This

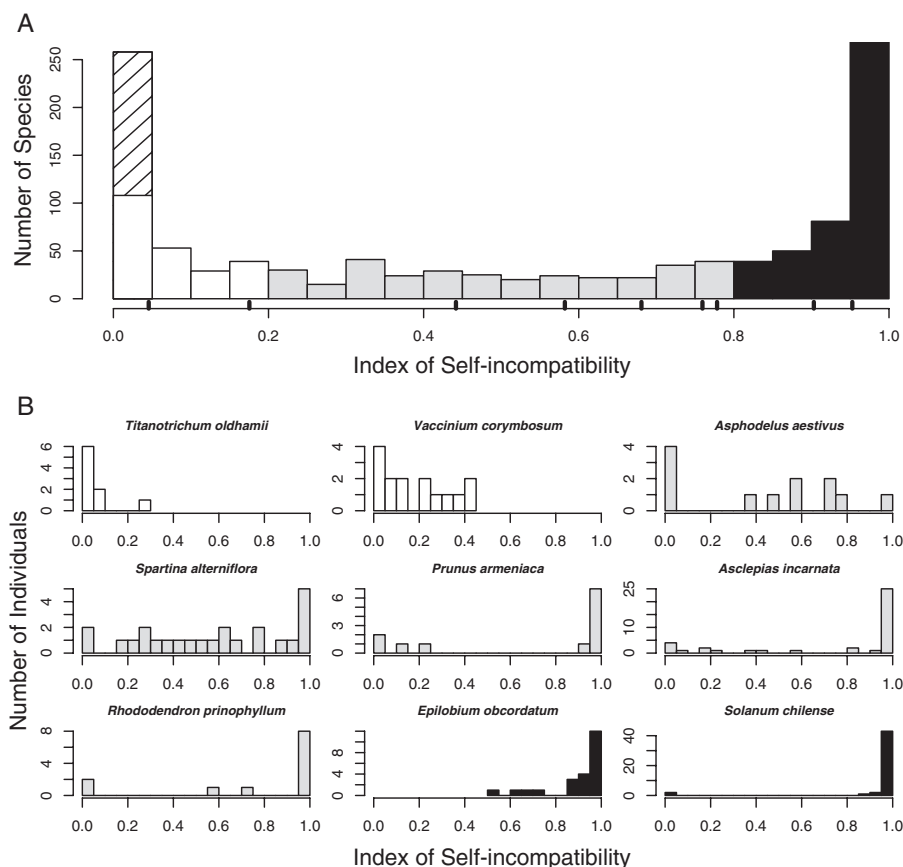


Figure 1. Histograms of ISI distributions reported in current study. ISI measures the strength of SI reaction by comparison of relative selfed and outcrossed success of fruit set per flower after manual pollinations. (A) Distribution of ISI reports collected from literature ($n = 1238$). Species with negative ISI values were set to zero (hatched area). (B) A selection of single-plant ISI distributions for species with at least eight reported values. Histograms are positioned of increasing species mean ISI by row. Species means for the single-plant ISI distributions are marked with notches on the ISI axis in panel A to further illustrate the effects of averaging.

estimate is in agreement with previous studies, and suggests that roughly one-half of extant flowering plants may be obligately outcrossing through SI or dioecy (Igić and Kohn 2006). Nearly three-quarters of species are aggregated at either end of the distribution of ISI means, although several complicating factors are on balance expected to yield an underestimate of bimodality.

DISTRIBUTION OF ISI VALUES

The inferred ISI distribution is shaped by both natural processes and experimental artifacts, subject to errors and biases that may overestimate the extent of bimodality. We discuss artifacts immediately below and then propose the likely causal processes involved, which could improve our understanding of the distribution of breeding systems. First, repeated measurement of ISI at several time points over the lifetime of an inflorescence or genet is time consuming and impractical, and a few exceptional studies record delayed selfing ability (transient SI; Good-Avila and Stephenson 2002; Goodwillie et al. 2004). As a consequence, practitioners may systematically choose younger flowers, near

pollen dehiscence, and fail to record widespread transient SI, or that the published studies are rare because they are only performed and successful when prior suspicions arise regarding the plausibility of transient SI. Furthermore, individuals with partial-SI, which also express strong early-acting inbreeding depression, may display $ISI \simeq 1$ (Kenrick and Knox 1989). The current dataset may also be biased with respect to its sampling of life history, morphological, geographic, or taxonomic characteristics. For instance, reporting may be skewed in favor of families with larger and individualized flowers and morphologies that are otherwise easily manipulated and scored in controlled pollination experiments.

It is difficult to establish the exact magnitude and direction of biases, but we expect that the extreme values of ISI may be instead underestimated in our data, especially at the high end of the distribution. Perhaps most trivially, because $ISI \leq 1$ (fruit set after selfing ≥ 0), many sources of error in scoring selfed fruit set can favor lower reported ISI values of truly SI individuals, away from $ISI = 1$. Errors may be common and substantial, especially in studies that perform small numbers of self- and cross-pollinations,

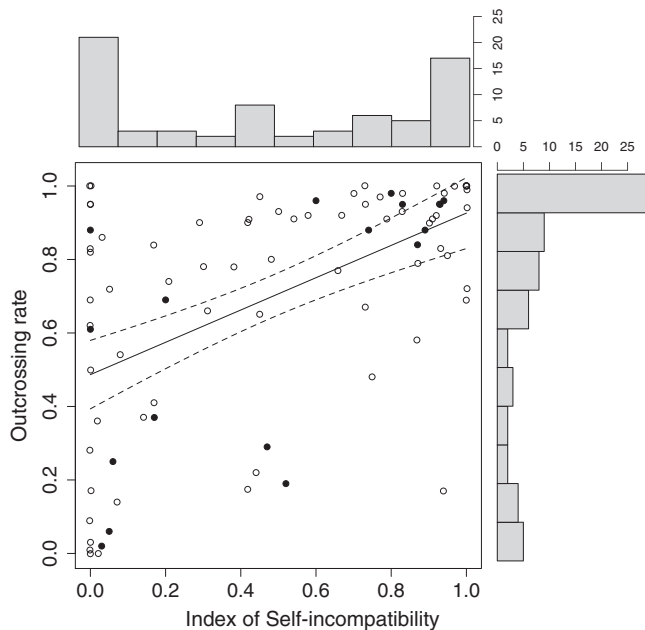


Figure 2. Correlation between ISI and outcrossing rate, t ($r^2 = 0.27$, $P < 0.001$, regression 95% confidence interval shown; see text for details). Outcrossing rate estimates are from Igić and Kohn (2006), Barringer (2007), and subsequently published individual reports. Filled black circles indicate instances where both measurements were taken from the same population. Histograms of ISI and outcrossing rate distributions are shown facing the respective labeled axes.

involving fewer than 20 flowers ($n = 108$). Occasionally, fruit set from natural pollinations is higher than that of either self- or cross-pollinations. One potential cause of the reduced fruit set in experimental crosses, especially in small populations, may be the use of pollen from incompatible donors, which are sourced from reproductively isolated individuals or share S-alleles with recipients.

The literature is replete with reports of seedless virgin fruits set through parthenocarpy. For example, in a recent study of the breeding system in *Kunzea pomifera* (Myrtaceae), Page et al. (2010) find substantial fruit set in unpollinated controls (44% in one plant), with no or low seed set. In *Styrax japonicus* (Styracaceae), Reed (2003) finds the mean fruit set declining from 22% one month after pollination to 0% two months later. Similar postpollination phenomena are known to occur widely (i.e., selective fruit abortion; Stephenson 1981), but it remains unclear to what extent they are adaptive. Also, many studies do not report controls for agamospermy. Each of these factors could lower ISI estimates—increase inferred fruit set after selfing—in otherwise SI species. A variety of reporting biases are also plausible, if not certain, and we find it reasonable to speculate that experimental confirmation of a high ISI for a species previously reported as SI could possibly receive relatively lower effort. Nevertheless,

the distribution of ISI reported here may approximate that of flowering plant species. It may even crudely reflect a long-term equilibrium, if one exists.

Studies in our ISI collection commonly report species values as averages of multiple individual or population measures. Even if individuals or populations are at the extremes of the distribution, but sampled in arbitrary ratios, the averaging effect would yield a misleading measure of species-wide ISI. It is possible that no individual or population in the sample exhibits the average ISI. Although not representative of the entire dataset, this is most clearly illustrated with our data from *Solanum chilense* (Solanaceae), whose individuals generally express a widespread form of gametophytic SI (Igić et al. 2007). Self-compatible mutants may be found segregating within nearly all large populations, thus decreasing the average species-wide ISI (0.92; Fig. 1). Both genic mutations and whole genome duplication can break down SI (the latter does so often in this and other species employing the RNase-based mechanism of SI), and each contributes to the presence of segregating factors that continually supply individuals with deleterious intermediate ISI values (Lewis 1943; Charlesworth 1985; Miller and Venable 2000; Robertson et al. 2011). In fact, the best available data indicates that SC mutations arise in SI plants at an appreciable rate, ranging from $\mu_{SC} = 2.0 \times 10^{-7}$ to 1.0×10^{-5} per pollen grain (de Nettancourt 1977; Lewis 1979). It is therefore not surprising that multiple independent SC mutants should be uncovered segregating at modest frequencies in otherwise mostly SI species (e.g., Ascher 1984), especially if they are partly recessive, associated with marginally lower fitness, or when they enjoy a transitory advantage. As a consequence, mutants may be eliminated at a highly varied pace, depending on the current genetic and ecological context, and yield disparate intraspecific ISI distributions. Observed ISI < 1 phenotypes may be recurrently generated and always comprise some fraction of individuals, at a fluctuating equilibrium hovering around a varying selection–mutation balance. Therefore, at least some intermediate ISI may persist in a sample of angiosperms, despite strong disruptive lower level selection. This suggests a valuable set of experiments quantifying the mutation rates and fitness consequences of the initial SC mutants in otherwise SI species. Variation in SI frequencies among different plant lineages could be partly determined by estimating such parameters.

Although some species averages are likely to be situated in the middle of the ISI distribution because of experimental and other artifacts, there is little doubt that some proportion of individuals, populations, and species exhibit genetically based intermediate ISI. Variability in the strength of SI has been recognized for well over a century (Stout 1938; Darwin 1876) and recently regained the spotlight at the center of a cohesive and influential set of empirical, theoretical, and synthetic studies in plant breeding system evolution (Stephenson et al. 2000;

Good-Avila and Stephenson 2002; Travers et al. 2004; Vallejo-Marin and Uyenoyama 2004; Goodwillie et al. 2005; Ferrer et al. 2009), which posit that it may represent a stable strategy and persist over evolutionary time. But there is still considerable disagreement over the dominant cause and longevity of the occurrence of intermediate ISI and, by extension, intermediate outcrossing rates.

WHAT SHAPES ISI DISTRIBUTION(S)?

The focus on the individual level in studies of plant mating and breeding system evolution advanced our understanding of the basic processes that govern their distributions. It also potentially mislead the interpretations of the distribution of mating and breeding systems, especially when coupled with a general disregard for speciation and extinction as potent drivers of evolution, whose effects can cascade to lower levels of evolutionary hierarchy. To facilitate discussion of this controversial topic, here we conflate “lower level” selection (individual level and below), in part because they take place on a shared time scale, the lifetime of individual plants. We also combine selection at higher levels (populations, species, and so on; here termed “species selection”), which occurs on time scales that are generally far greater than individual plant lifetimes.

Nearly all theoretical studies of breeding system evolution to date have modeled only lower level phenomena and, with this limitation in mind, some support the idea that a range of equilibrium species-wide average ISI values is possible, if not likely (Vallejo-Marin and Uyenoyama 2004; Porcher and Lande 2005). For example, these studies find that SC mutations may be maintained at arbitrary fractions within a plausible range of parameters (Porcher and Lande 2005), and partial SI can exist at least in models with constant inbreeding depression (Vallejo-Marin and Uyenoyama 2004). But these modeling approaches fail to allow species to duplicate or become extinct at any rate, despite the fact that differential speciation and extinction rates can yield arbitrary equilibria in absence of stabilizing lower level selection (Igić and Kohn 2006). Moreover, a recent empirical study suggests a substantial role for species selection in shaping breeding system evolution (Goldberg et al. 2010).

Selection at the higher levels of evolutionary hierarchy may be a major force shaping the distribution of breeding systems (Dobzhansky 1950; Stebbins 1957; Williams 1992; Holsinger 2000). Species-wide mean ISI is an aggregate trait inherited by descendent lineages, and one that displays large interspecies variation. Furthermore, it is widely acknowledged that breeding systems may strongly influence the differential fitness of species, their net diversification rates, through the process sometimes termed “effect macroevolution” (Stebbins 1957; Lande and Schemske 1985; Takebayashi and Morrell 2001; Igić et al. 2006; Schoen and Busch 2008; Jablonski 2008; Goldberg et al. 2010). Both theoretical and empirical studies of plant breeding systems have largely neglected to consider the action of higher level

selection, which is expected to strongly oppose selection at the lower levels of evolutionary hierarchy (Igić et al. 2008; Goldberg et al. 2010). This expectation was held in high regard even by the most ardent critics of the broad importance of higher level selection (Williams 1992, p. 36). Additionally, it is compatible with the view that mating and breeding systems are commonly shaped by long-term stabilizing selection favoring mixed strategies. For example, if inbreeding depression is fixed, or purged very slowly, then a mixed breeding strategy of plants with intermediate ISI could easily become stable in a population or species (e.g., Vallejo-Marin and Uyenoyama 2004). But if mixed breeding systems are associated with a strongly negative net diversification rate, selection at the lower level can fix an allele encoding a mixed strategy, a trait that causes species to diversify at a slower or negative rate. Depending on the strength and direction of selection at various levels of hierarchy, this process could result in an arbitrarily low proportion of populations employing mixed breeding strategies. Analogously, if mixed mating is occasionally maintained by stabilizing selection driven by, for instance, correlated fertility components (Johnston et al. 2009), association of mixed mating and low diversification rates could still potentially result in a bimodal distribution of mating systems.

Schoen and Busch (2008) and Igić et al. (2008) simultaneously introduced variations on a simple model to describe the dynamics of reproductive evolution, which we modify below to further illustrate its significant implications for the distribution of breeding systems. The principal aim is to demonstrate that a null hypothesis positing disruptive lower level selection and higher diversification rate of SI species as the primary drivers of breeding system evolution, is consistent with a vast range of ISI distributions, including the one we find in our study. We simplify the distribution of breeding systems by arbitrarily classifying it into three states: SI ($ISI \geq 0.8$), “partial” SI (pSI; $0.2 < ISI < 0.8$), and SC ($ISI \leq 0.2$). The lineages in each of the three states can diversify with a state-specific balance of speciation and extinction rate (r_I , r_P , and r_C , respectively). Furthermore, for computational simplicity, we fix the rate of loss of SI (q_{IP}) and the rate of transition from pSI to SC (q_{PC}) to be equal, $q_{IP} \approx q_{PC}$, although this assumption could easily be relaxed. The magnitude of q_{IP} may be so much greater than the gain of SI, so as to be nearly irreversible (Igić et al. 2008), and q_{PC} is expected to have similar dynamics—driven by a positive feedback between modifiers of outcrossing rate and inbreeding depression (Bull and Charnov 1985; Lande and Schemske 1985). We consequently disallow reversals, making a further assumption not crucial to the point at hand, but greatly simplifying calculations. Specifically, the argument is approximately correct so long as the transition asymmetry is very high (≈ 100 -fold). As a result, the shape of the ISI distribution is then determined by the above set of two transition rates and three net diversification rates.

The frequency of each ISI “state” reported in this study is 43.9% SI, 25.5% pSI, and 30.6% SC. The present ISI distribution may be recovered if each state-specific diversification rate is $r_I = 0.43$ and $r_P = r_C = -0.18$ [lineage⁻¹ my⁻¹], and species transition unidirectionally between states at an equal rate, $q_{IP} = q_{PC} = 0.34$ (eq. 1 in Igić et al. 2008). Different model parameters can yield a variety of distributions. Measuring them could shed light on disparity in ISI between annuals and perennials, wind- and animal-pollinated, or woody and herbaceous species. Also, this shows that the presence of species with intermediate ISI does not demonstrate their long-term evolutionary stability. Species with intermediate ISI may be transitioning along an irreversible path toward lower ISI, or instead headed toward extinction. Moreover, the rate at which alleles that modify the expression of SI increase in frequency is unclear, as is the ability of stabilizing low-level selection to produce partially SI species. It is difficult to experimentally demonstrate that stabilizing low-level selection maintains intermediate ISI over many generations. Nevertheless, ignoring the action of species selection probably reduces our ability to accurately capture the evolutionary dynamics of breeding systems and explain their frequency distribution (Igić et al. 2008; Goldberg et al. 2010). We do not hold that the lower and higher level perspectives and approaches to the understanding of breeding system evolution are in opposition. Instead, we suggest that they each (alone) represent a partial perspective of a complete research program, which can lead to misleading conclusions.

ISI AND OUTCROSSING RATE

The binary classification (presence or absence) of SI may affect the assumed average outcrossing rates of species. Conventionally, species that are classified as SI are considered, along with dioecious species, to be obligately outcrossing (e.g., Silvertown et al. 2002). This assumption may result in an underrepresentation of SI species in the outcrossing rate literature due to biases against measuring the outcrossing rate of SI species (Igić and Kohn 2006). We find that species with high ISI generally have high outcrossing rates, as expected. In contrast, SC species may exhibit a number of breeding system characteristics to achieve cross-pollination, and we expect species with low ISI to have a wide range of outcrossing rates. The relationship between ISI and outcrossing rate collected in the present study shows a positive correlation ($r^2 = 0.27$, $P < 0.001$), with significantly more variation found in species with $ISI \leq 0.2$ (Fligner–Killeen test: med $\chi^2 = 16.97$, $P < 0.05$) (Fig. 2). Considerable variation is still found in the outcrossing rates of species with $ISI \geq 0.8$. The departure from the expected relationship between ISI and t in the subset of species with $ISI \geq 0.8$ may be explained by the disparities in the manner in which the dataset was assembled. ISI and t measurements were obtained from different studies for the

majority of species, and experiments were often performed on populations across a species’ range and at different times.

An instructive example of the potential influence of temporal and spatial setting on ISI and t measurements comes from the Caribbean red mangrove *Rhizophora mangle* (Rhizophoraceae), which stands out in our dataset as the lone datapoint with high ISI and low t . ISI was measured in Venezuelan populations ($ISI = 0.94$; Lemus-Jimenez and Ramirez 2005) and outcrossing rate was estimated in Puerto Rico ($t = 0.17$; Klekowski et al. 1994). It is entirely possible that the Puerto Rican taxon belongs to a different species, or that it is a relatively isolated SC population. Ideally, comparisons of breeding and mating system characteristics should involve measurements from the same individual collected at the same time, perhaps over time and across the distribution of species. To our knowledge, however, no such results have been published, although appropriate studies may have been performed (see, e.g., Foxe et al. 2010).

In this article, we examined the distribution of ISI, its implications and causes, using the most comprehensive dataset collected to date. Despite numerous factors that are expected to make the collected data conservative, three-quarters of all species are found at the extremes of the distribution. We propose that a simple null model explains the presence of species and individuals displaying intermediate ISI values without invoking long-term stability of partial-SI mutations. Careful study of the strength of selection at different levels of evolutionary hierarchy is necessary for strong conclusions regarding the causes of the distribution of breeding systems.

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Supporting Information

The following supporting information is available for this article:

Figure S1: Histogram of unmodified ISI_{fruit} distribution in current study (note the break in the abscissa).

Figure S2: Histograms of ISI_{fruit} distribution from the most common families in current study.

Figure S3: Histogram of ISI_{fruit} distribution excluding data from the most common families ($n = 679$).

Figure S4: Histogram of ISI_{seed} distribution from current study ($n = 487$). ISI_{seed} data was excluded from the main analyses because of unknown effects of non-SI related phenomena on seed per fruit occurrence.

Figure S5: Histogram of the incidence of agamospermy from reports that specifically measured its occurrence ($n = 271$).

Figure S6: ISI_{fruit} distributions of various discrete characters. In order to assess how ISI is correlated with other traits, each species was classed as either woody or herbaceous, annual or perennial, and animal or wind pollinated when data was available.

Table S1: List of all species and source studies.

Supporting Information may be found in the online version of this article.

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